A

Research Plan

On

SOVEREIGN AGENTIC COMMERCE: DESIGNING EDGE-DEPLOYED MULTI-AGENT SYSTEMS FOR INDIA'S OPEN NETWORK FOR DIGITAL COMMERCE (ONDC)

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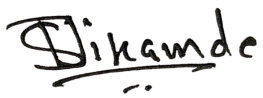
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DECLARATION

I hereby affirm that the work **"Sovereign Agentic Commerce: Designing Edge-Deployed Multi-Agent Systems for India's Open Network for Digital Commerce (ONDC)"** presented in this synopsis is exclusively my own & there are no collaborators. It does not contain any work for which a degree/diploma has been awarded by any other University/Institution. All the ideas and references have been duly acknowledged.



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CERTIFICATE

This is to certify that the research work **"Sovereign Agentic Commerce: Designing Edge-Deployed Multi-Agent Systems for India's Open Network for Digital Commerce (ONDC)"** allotted to Dr. Himagi Verma would be a beneficial research work in the field of Computer Science & Applications. This would be a new research work and would be within the scientific framework of research.

Signature of Supervisor……

PREFACE

This handbook presents the guidelines for Ph.D Scholars to write synopsis of research and thesis/dissertations. These guidelines will be very helpful both to the students and their Supervisors for preparation of their manuscripts and also brought uniformity to the whole process. The previous guidelines are revised as per the valuable suggestions from the Desh Bhagat University members. Further suggestions are always welcomed. These instructions will help in saving a lot of time for both- scholars and committee for the preparation and the approval of the synopsis.

SOVEREIGN AGENTIC COMMERCE: DESIGNING EDGE-DEPLOYED MULTI-AGENT SYSTEMS FOR INDIA'S OPEN NETWORK FOR DIGITAL COMMERCE (ONDC)

1. INTRODUCTION

The Algorithmic Transformation of Digital Public Infrastructure. The global digital economy is experiencing a significant structural reconfiguration, transitioning from centralised, monopolistic platforms to decentralised, interoperable protocols. This transition is particularly evident in emerging economies such as India, which has effectively spearheaded the notion of Digital Public Infrastructure (DPI) on a national scale. Building on the fundamental success of the Aadhaar biometric identity system and the Unified Payments Interface (UPI) for efficient financial transactions, the Government of India introduced the Open Network for Digital Commerce (ONDC) to democratise digital commerce. ONDC utilises the open-source Beckn protocol, which disaggregates the e-commerce value chain, enabling buyers and sellers from disparate, competing apps to discover, negotiate, and deal with each other. In this manner, ONDC systematically eliminates the elevated commission structures of traditional platforms (often ranging from 15% to 35%) and lowers them to a more sustainable 3% to 12%.

1.1 The Operational Paradox Concerning Micro, Small, and Medium Enterprises (MSMEs)

Although ONDC successfully removes financial obstacles for Micro, Small, and Medium Enterprises (MSMEs), it unintentionally creates a significant operational contradiction. In a centralized marketplace, such as Amazon, the platform algorithmically manages search visibility, dynamic price, logistical routing, and consumer complaint resolution. In a decentralised network, the responsibility for supply chain orchestration and real-time API management is placed squarely on the individual MSME. Conventional small business proprietors, possessing low digital proficiency and restricted resources, must now manoeuvre through intricate, asynchronous digital landscapes. This cognitive and operational burden constitutes a substantial obstacle to the extensive implementation of decentralised trade.

The Transition to Agentic Commerce To address this friction, the technological paradigm is swiftly evolving from reactive Generative AI to "Agentic AI." Unlike traditional Large Language Models (LLMs) that provide passive text outputs upon request, Agentic AI systems are assigned a high-level objective, autonomously devise intermediate steps, utilise external software tools, and conduct transactions across decentralised networks without human involvement. Projections suggest that AI agents could facilitate between $3 trillion and $5 trillion of global consumer commerce by 2030. McKinsey's 2025 State of AI report indicates that 62% of organisations are

now experimenting with AI agents, while 23% are actively scaling agentic systems in at least one business function.

This change is facilitating the "Agentic Web" an ecosystem in which autonomous software agents mediate digital interactions. Projects like the MIT Media Lab's Project NANDA (Networked AI Agents in Decentralised Architecture) are constructing the essential infrastructure necessary for billions of AI agents to identify, authenticate, and organise jobs inside open networks. The launch of the Model Context Protocol (MCP) by Anthropic in late 2024, currently overseen by the Linux Foundation's Agentic AI Foundation, has established a worldwide open standard for securely connecting AI agents to various data sources, significantly enhancing their operational efficacy.

Nevertheless, initial implementations of agentic commerce on the Beckn protocol depend exclusively on centralized, proprietary cloud systems such as OpenAI's ChatGPT. This dependence results in exorbitant "pay-per-query" infrastructure expenses, poses significant data privacy risks, and marginalises rural populations due to the English-centric orientation of foundational models. The RAND Corporation has determined that the deployment of agents on fragmented, centralised data is the primary cause of failures in agentic AI projects.

1.2 The Necessity for Sovereign AI and Edge Computing

A conclusive, scalable approach necessitates the building of a "Agentic DPI" implementing highly optimised Small Language Models (SLMs) directly at the edge. Executing these models on local hardware minimises inference costs to only electricity expenses, while ensuring stringent data sovereignty. Moreover, to address the digital divide, these edge architectures must inherently incorporate sovereign language translation models, such as India's government-supported Bhashini platform. Designing this localised, vernacular, multi-agent workflow is the essential next step in democratising the Indian digital economy.

1.3 Articulation of the Issue / Research Inquiry

This research addresses the primary issue of operational friction encountered by MSMEs in shifting to decentralised digital commerce networks (ONDC), exacerbated by architectural constraints, expenses, and language hurdles associated with existing cloud-dependent AI assistants.

The Beckn protocol decentralises commerce using intricate, asynchronous Application Programming Interfaces (APIs) - notably through paired callbacks like search and on_search, or select and on_select. The management of real-time data streams imposes an untenable cognitive burden on small vendors. Efforts to automate this orchestration through centralised Large Language Models present three significant systemic bottlenecks:

- **Economic Unsustainability:** Elevated and recurrent API inference expenses linked to proprietary cloud LLMs undermine the margin advantages achieved by transitioning to ONDC.
- **Data Privacy Risks:** Transmitting sensitive proprietary business inventories, pricing tactics, and customer data to third-party offshore cloud providers fundamentally contravenes data sovereignty principles.
- **Algorithmic Unreliability:** LLMs are significantly susceptible to fabricating the stringent JSON schema payloads mandated by the Beckn protocol, resulting in disastrous transaction failures.

A significant operational and technological impediment exists: there is no rigorously validated, open-source architectural framework for implementing localised, edge-based multi-agent systems capable of autonomously executing e-commerce operations in local Indian vernaculars without dependence on proprietary cloud infrastructure or encountering failures in rigid API schema generation.

2. REVIEW OF LITERATURE

The incorporation of artificial intelligence into decentralised network protocols requires a simultaneous advancement in edge computing, natural language processing, and system orchestration. A careful analysis of current literature underscores significant progress and notable constraints in this nascent field.

The Evolution of Agentic AI and Decentralised Commerce Recent scholarship extensively documents the transition from passive AI to active, goal-directed agentic systems. The five most significant e-commerce applications for agentic AI currently identified are inventory management, dynamic pricing, customer service resolution, hyper-personalization, and fraud detection. Studies have shown that integrating Agentic AI with decentralised protocols such as ONDC can reduce decision-making latency from days to mere seconds, significantly enhancing a business's operational agility. However, contemporary literature often presumes the existence of large, centralised computing clusters to support these agents.

2.1 Protocol-Level Automation, MCP, and Project NANDA

The notion of the "Agentic Web" is acquiring considerable scholarly attention. Project NANDA at the MIT Media Lab proposes an architecture wherein a minimal index resolves to dynamic, cryptographically verifiable "AgentFacts," facilitating privacy-preserving discovery across organisational borders through a CRDT-based update protocol. The industry is concurrently standardising on the Model Context Protocol (MCP). Created by Anthropic and contributed to

the Agentic AI Foundation in December 2025, MCP offers a standardised interface for file reading and function execution, addressing the fragmentation that previously affected agentic architectures. Additionally, 2026 has witnessed the introduction of the Universal Commerce Protocol (UCP), which creates a standardised language for agents throughout the purchasing experience and maintains compatibility with MCP.

Sovereign AI, Vernacular Inclusion, and Bhashini necessitate linguistic inclusivity in the democratisation of digital commerce. English-centric AI models systematically marginalise the majority of the Indian populace. Bhashini, India's national public digital infrastructure, has undergone considerable evolution; recent literature highlights its transition from basic multilingual access to "intersectional intelligence," integrating language comprehension with specialised domain knowledge in areas such as agriculture and commerce. Recent implementations on ONDC, shown by the Bhashini-enabled "Sahayak" bot, effectively facilitate conversational commerce in five different Indian languages. Nonetheless, the integration of unstructured colloquial voice searches into the intricate, multi-step structured API calls mandated by ONDC at the edge is still a predominantly uncharted technical domain.

2.2 Grammar-Constrained Decoding for API Dependability

A significant constraint observed in the implementation of LLMs for software automation is their failure to reliably provide syntactically valid structured outputs, such as JSON. Recent computational research has established "Grammar-Constrained Decoding" (GCD) to overcome this issue. Methods like XGrammar convert schemas into Finite-State Machines (FSMs), thereby eliminating faulty tokens during the generation phase of the LLM. In 2026, researchers at ICLR unveiled Parser Stack Classification (PSC), an innovative GCD approach that generates masks by examining the parser stack precisely once during each decoding step, attaining JSON schema production speeds up to 30 times quicker than prior benchmarks. Although GCD demonstrates significant efficacy in controlled benchmarks, its implementation in an asynchronous, multi-agent Beckn context with localised edge models necessitates more empirical validation.

3. RESEARCH GAP AND PROBLEM FORMULATION

A comprehensive analysis of the existing literature in computer science and economics identifies four specific academic and operational gaps.

3.1 The Edge-Orchestration Gap: Present autonomous commerce frameworks predominantly rely on proprietary cloud-based LLMs and cloud-based MCP implementations. No documented, peer-reviewed architectural framework exists for multi-agent orchestration employing localised, open-weight Small Language Models (SLMs) deployed solely on limited edge hardware.

3.2 The Asynchronous Latency Gap: The Beckn protocol is fundamentally dependent on asynchronous webhooks (on_search, on_select, on_confirm). The literature is deficient in comprehensive programmatic frameworks to address the latency and synchronisation issues that arise when autonomous AI agents are required to await, process, and respond to delayed network callbacks.

3.3 The Vernacular Intent Gap: Despite the existence of Bhashini's "intersectional intelligence" APIs, there is a significant deficiency in comprehensive frameworks capable of precisely translating unstructured, colloquial rural voice queries into the intricate semantic intent vectors necessary for facilitating accurate ONDC product discovery without human intervention.

3.4 The Structured Output Gap: Research indicates that smaller edge models face considerable challenges in maintaining strict schema compliance. Empirical studies utilising advanced 2026 Grammar-Constrained Decoding (GCD) methods, such as Parser Stack Classification (PSC), are lacking in their application to Beckn Protocol JSON payloads to ensure defect-free API creation.

4. OBJECTIVES OF THE STUDY

This study is necessitated by the critical convergence of decentralised digital economics and the constraints of existing artificial intelligence applications. The study specifically examines:

Financial Inclusion and Margin Protection: The necessity to entirely eradicate "pay-per-query" cloud AI inference expenses. For MSMEs functioning on minimal margins, the automation of activities must not incur additional, unsustainable SaaS expenses.

Data Sovereignty and Privacy: The necessity to safeguard merchant and consumer information. An entirely localised, edge-deployed architecture ensures that important inventory and customer data remains on the owner's local hardware, hence reducing the dangers of data scraping.

Operational Autonomy for MSMEs: The necessity to furnish small business proprietors with an independent "virtual workforce" adept at managing dynamic pricing, ongoing inventory monitoring, and optimal logistics routing in real time.

Addressing the Digital Divide: The necessity to dismantle digital literacy obstacles by developing systems that inherently comprehend and process local Indian vernacular languages via intuitive voice interfaces, hence expanding the digital economy to Tier-3 cities.

4.1 Goals

The main aim of this research is to design, implement, and quantitatively assess a privacy-preserving, localised multi-agent Small Language Model (SLM) architecture that can independently perform end-to-end e-commerce workflows on the decentralised Beckn protocol (ONDC), without dependence on centralised cloud APIs.

To methodically attain this principal purpose, the study will pursue the subsequent specific sub-objectives:

- **Perception Agent Architecture:** To develop and implement an edge-deployed "Perception Agent" that use Bhashini models to effectively analyse unstructured vernacular queries and convert them into dense semantic intent vectors.
- **Reasoning and Logistics Orchestration:** To develop an autonomous multi-agent pipeline proficient in efficiently handling the intricate asynchronous callback states characteristic of the Beckn Application Protocol (BAP), utilising standards similar to the Model Context Protocol (MCP).
- **Implementation of Schema Enforcement:** To incorporate and assess advanced Grammar-Constrained Decoding (GCD) techniques, particularly focusing on Parser Stack Classification (PSC), within the SLM generation pipeline to ensure the syntactic accuracy of Beckn JSON payloads through mathematical validation.
- **Quantitative Validation:** To systematically evaluate the localised edge system's Task Completion Rate (TCR), decision latency, and semantic alignment in comparison to established cloud-dependent benchmarks.

4.2 Extent

This research is precisely defined to guarantee academic rigour and practical applicability at the convergence of Multi-Agent Systems, Edge Computing, and Decentralised Networks.

Domain Scope: The research is only confined to Business-to-Business (B2B) and Business-to-Consumer (B2C) retail digital commerce and logistics operations utilising the Open Network for Digital Commerce (ONDC).

The study examines the implementation of heavily quantised, open-weight SLMs (e.g., Llama-3 8B, Mistral) on edge hardware configurations appropriate for MSMEs. It employs the Beckn Application Protocol (BAP), Model Context Protocol (MCP) standards, and Bhashini APIs.

The orchestration of agents will concentrate on the precise sequence of discovery, selection, initialisation, and confirmation within the e-commerce transaction lifecycle.

The evaluation scope encompasses the mathematical quantification of system performance through processing latency (expressed in milliseconds), intent matching accuracy, and operational Task Completion Rates (TCR).

4.3 Study Delimitation

This research carefully delineates the following delimitation to preserve academic rigour and maintain viability within a conventional PhD timeline:

1. **No Foundational Training:** The research will not endeavour to develop a new foundational Large Language Model from the ground up. It will exclusively employ quick engineering, fine-tuning methodologies (including LoRA), and grammar-constrained decoding on pre-existing open-weight models.

2. **Simulation Environment:** Testing, execution, and validation will occur solely within the safe, open-source Beckn Developer Sandbox. The experimental agents will not be implemented in the operational, commercial ONDC financial network.

3. **Sector Limitation:** Although the Beckn protocol is applicable across various sectors, this study is confined to the e-commerce and logistics fields. It omits other nascent Beckn applications, like healthcare and mobility.

4. **Hardware Limitations:** The assessment of edge deployment will be conducted on consumer-grade to high-end workstation hardware, emulating MSME capabilities, explicitly excluding dependence on huge enterprise data center GPU clusters.

4.4 Investigative Enquiries

This study aims to address the following fundamental scientific questions:

RQ1: How can a localised, multi-agent SLM architecture be systematically designed to address asynchronous callback latency and intricate state management challenges in decentralised Beckn networks, while avoiding systemic transaction failures?

RQ2: To what degree can the incorporation of localised vernacular models, such as Bhashini's intersectional intelligence API, at the perception layer enhance the semantic intent matching of rural e-commerce enquiries in comparison to conventional search methods?

RQ3: Does the application of sophisticated Grammar-Constrained Decoding (e.g., Parser Stack Classification) on localised SLMs substantially diminish the incidence of API schema hallucinations and faulty JSON payloads in comparison to unconstrained generative models?

RQ4: Is it possible for a fully edge-deployed, autonomous workflow to attain Task Completion Rates (TCR) and decision delay metrics that are statistically equivalent to cloud-dependent models, while concurrently eliminating external inference costs?

5. RESEARCH HYPOTHESES

In light of the recognised gaps in the literature and the established research issues, the study proposes the following hypotheses:

Null Hypothesis 1: Edge-deployed multi-agent systems demonstrate a markedly reduced Task Completion Rate (TCR) in processing ONDC transactions relative to centralized, cloud-based LLM designs, attributable to the limited reasoning capabilities of smaller models.

Alternate Hypothesis 1: The integration of Grammar-Constrained Decoding with stringent timely orchestration will enable an optimised edge-deployed multi-agent SLM system to attain a statistically comparable Task Completion Rate to cloud baselines.

Null Hypothesis 2: The incorporation of vernacular voice translation models results in an increased semantic error rate that hinders the precise creation of JSON payloads for the Beckn protocol.

Alternate Hypothesis 2: Integrating Bhashini intent models into a specialised Perception Agent guarantees precise, machine-readable API payload generation, markedly decreasing the transaction abandonment rate for non-English customers.

6. RESEARCH METHODOLOGY

This study utilises a quantitative, computational systems engineering approach. The emphasis is on iterative development, local deployment, and stringent mathematical benchmarking of the proposed multi-agent architecture within a regulated network simulation.

6.1 Approach

The research employs an experimental computational approach. It entails a multi-stage software engineering methodology:

1. **Architectural Phase:** Designing the topology of the multi-agent system, encompassing Perception, Reasoning, and Logistics agents.

2. **Implementation Phase:** Containerising the Beckn protocol with Beckn-ONIX and installing localised SLMs utilising frameworks such as Ollama.

3. **Orchestration Phase:** Configuring the communication middleware to manage asynchronous Webhooks.

4. **Evaluation Phase:** Conducting simulated transactions and quantitatively recording performance data relative to a cloud baseline.

6.2 Field of Study

The research functions at the intricate convergence of Artificial Intelligence (Agentic Workflows and Natural Language Processing), Distributed Systems (Decentralised Network Protocols), and Edge Computing.

6.3 Specimen and Surroundings

The research will depend on highly precise digital twin simulations to prevent disruption of live banking networks.

Environment: The official ONDC/Beckn Sandbox environments are deployed locally.

Sample Data: A created dataset of 5,000 extremely different simulated B2B and B2C commercial enquiries, encompassing straightforward direct requests as well as intricate, multi-variable logistical queries in several vernacular languages.

6.4 Methods

The study will employ sophisticated computer science methodologies to address the highlighted deficiencies:

- **Multi-Agent Orchestration:** A systematic linkage wherein a "Perception Agent" processes unstructured data, a "Reasoning Agent" manages decision-making logic, and a "Logistics Agent" assesses best delivery routes.
- **Grammar-Constrained Decoding (GCD):** Employing sophisticated parsers (such as XGrammar or PSC) directly on the SLM output generation to specifically prohibit tokens that contravene the stringent JSON syntax mandated by Beckn APIs.
- **Semantic Intent Vectorisation:** Employing dense vector embeddings to align unstructured vernacular enquiries with MSME product catalogues.
- **Asynchronous State Management:** Creating localised memory buffers to save the state of a transaction while network agents await delayed webhooks.

6.5 Instruments

The experimental infrastructure will be developed utilising open-source, sovereignty-preserving technology.

- **Network Protocol:** Beckn-ONIX plugin architecture for localised network implementation.
- **AI Deployment:** The Ollama framework and vLLM for serving quantised, open-weight language models exclusively on local edge hardware.
- **Integration Protocols:** Execution of the Model Context Protocol (MCP) to standardise the utilisation of agent tools.
- **Vernacular Processing:** Bhashini API environments.
- **Agent Frameworks:** LangChain or AutoGen.

6.6 Measurement and Scaling

The study employs precise, measurable indicators to guarantee stringent academic validation:

Task Completion Rate (TCR): The essential percentage of successfully executed end-to-end simulated orders devoid of human participation.

Payload Validity Rate (PVR): The proportion of generated JSON API calls that successfully meet stringent Beckn schema validation on the initial try.

Decision Latency: The median processing duration (in milliseconds) between the receipt of user input to the transmission of the API payload across the network.

Cost Efficiency Ratio: A mathematical measurement of the financial savings realised by the removal of cloud API inference charges.

6.7 Data Processing

The initial data processing entails the creation of the vector database for the "virtual" MSME. Simulated product catalogues will be assimilated and converted into highly dimensional vector embeddings. Incoming queries, handled by Bhashini, will undergo tokenisation, embedding, and cosine similarity searches to determine best product matches prior to being forwarded to the Reasoning Agent.

6.8 Evaluation and Appraisal

The analysis step will systematically assess the logs produced by the Beckn Sandbox. The system will monitor each transaction phase, pinpointing precisely where drop-offs or timeouts transpire. The failure sites will be examined to assess the reliability of edge SLMs in comparison to cloud LLMs.

6.9 Evaluating Hypothesis

The proposed hypotheses will be assessed utilising formal statistical techniques. The TCR and decision latency of the experimental Edge-Agentic system will be evaluated against a control group. An analysis of variance (ANOVA) or paired t-tests will be employed to ascertain the statistical significance of the data. If the proposed system sustains a high TCR ($\geq 90\%$) with a P-value, the null hypothesis will be rejected.

6.10 Anticipated Results

The successful execution of this research is expected to produce significant results:

An open-source "Agentic Edge Blueprint": A comprehensive, proven software architecture employing MCP and GCD, enabling developers to build autonomous agents for MSMEs.

Empirical Validation: Mathematical confirmation that highly quantised Small Language Models can effectively traverse the intricacies of decentralised digital commerce networks.

Democratisation of E-commerce: The establishment of an operational framework enabling non-English speaking, rural entrepreneurs to navigate intricate digital supply chains solely through voice commands.

6.11 Significance of Anticipated Outcomes

The findings of this study have significant significance for both computer science academics and global macroeconomic policy.

This research addresses a significant gap in distributed AI systems. It shifts the conversation from singular LLMs to efficient, tightly regulated multi-agent networks functioning at the edge.

This research offers a framework for genuine operational autonomy in the Indian MSME sector, enabling small enterprises to leverage independent automation without financial dependence on global cloud AI monopolies.

6.12 Proposed Plan Of Work

The planned research would be conducted as below during the PhD period:

Academic Year	Aug 25 - June 26				July 26 - June 27				July 27 - June 28			
Quarters	1	2	3	4	1	2	3	4	1	2	3	4
Coursework												
Literature Review Problem defination & Synopsis Submission												
Methodology Development & Initial Experiments												
Data Collection & Core Experiments												
Advanced Analysis & Validation												
Synthesis & Initial												

Drafting												
Final Review & Refinement												

Months 1 - 6: Comprehensive literature review concentrating on Agentic AI, Beckn protocol specifications, MCP architectures, and GCD parsing techniques.

Months 7 - 12: Installation and configuration of the Beckn-ONIX local sandbox employing Docker.

Months 13 - 18: Targeted implementation of open-weight SLMs utilising Ollama. Developing the Perception Agent to incorporate Bhashini translation functionalities.

Months 19 - 24: Development of the programmatic middleware connecting the agents to the Beckn Application Protocol (BAP). Execution of Grammar-Constrained Decoding (GCD).

Months 25 to 30: Comprehensive simulation executions within the sandbox environment. Meticulous mathematical documentation of TCR, PVR, and Decision Latency.

Months 31 - 36: Data synthesis, thesis finalisation, and preparation for the doctoral defence.

6.13 Provisional Chapter Outline

The final thesis will be organised to distinctly delineate the theoretical foundation, architectural engineering, and mathematical assessment of the research.

Chapter 1: Introduction: Outlines the macroeconomic transition towards Decentralised Commerce (ONDC) and presents the notion of Sovereign Agentic AI.

Chapter 2: Literature Review: A critical examination of existing studies on the Beckn Protocol, Project NANDA, Model Context Protocol (MCP), and the advancement of edge-based AI.

Chapter 3: Proposed System Architecture and Methodology: Outlines the comprehensive research framework and the quantitative assessment measures.

Chapter 4: Development of Vernacular Perception Agents: Concentrates on the incorporation of Bhashini APIs, parsing of unstructured voice queries, and vectorisation of semantic intents.

Chapter 5: Multi-Agent Orchestration and Schema Enforcement: The foundational engineering component of the thesis. Elucidates the administration of asynchronous Beckn webhooks and the particular execution of Grammar-Constrained Decoding (GCD).

Chapter 6: Results and Performance Analysis: Provides a quantitative validation of the framework, examining TCR, JSON payload accuracy, and latency.

Chapter 7: Conclusion and Future Scope: Integrates the commercial and scholarly ramifications of the Edge-Agentic framework.

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